

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$C_{\alpha\beta\chi}$	$C_{\chi\beta\alpha} == C_{\alpha\beta\chi}$	$-\frac{1}{2} C_{2^{-}\alpha\beta\chi}^{\#1} + \frac{1}{2} C_{2^{-}\beta\chi\alpha}^{\#1} + C_{3^{-}\alpha\beta\chi}^{\#1} - \frac{1}{6} \eta_{\beta\chi} C_{1^{-}\alpha}^{\#4} + \frac{1}{3} \eta_{\alpha\chi} C_{1^{-}\beta}^{\#4} - \frac{1}{6} \eta_{\alpha\beta} C_{1^{-}\chi}^{\#4} + \frac{1}{15} \eta_{\beta\chi} C_{1^{-}\alpha}^{\#2} + \frac{1}{15} \eta_{\alpha\chi} C_{1^{-}\beta}^{\#2} +$ $\frac{1}{15} \eta_{\alpha\beta} C_{1^{-}\chi}^{\#2} + \frac{1}{9} \eta_{\beta\chi} C_{0^{+}n_{\alpha}}^{\#2} - \frac{1}{2} C_{1^{+}\beta\chi}^{\#1} n_{\alpha} + \frac{1}{3} C_{2^{+}\beta\chi}^{\#2} n_{\alpha} - \frac{1}{3} C_{2^{+}\beta\chi}^{\#2} n_{\alpha} - \frac{1}{9} \eta_{\beta\chi} C_{0^{+}n_{\alpha}}^{\#3} + \frac{1}{9} \eta_{\alpha\chi} C_{0^{+}n_{\beta}}^{\#2} + \frac{1}{3} C_{2^{+}\alpha\chi}^{\#1} n_{\beta} +$ $\frac{2}{3} C_{2^{+}\alpha\chi}^{\#2} n_{\beta} + \frac{2}{9} \eta_{\alpha\chi} C_{0^{+}n_{\beta}}^{\#3} + \frac{1}{6} C_{1^{-}\chi}^{\#4} n_{\alpha} n_{\beta} - \frac{1}{15} C_{1^{-}\chi}^{\#2} n_{\alpha} n_{\beta} - \frac{1}{3} C_{1^{-}\chi}^{\#3} n_{\alpha} n_{\beta} + \frac{1}{3} C_{1^{-}\chi}^{\#1} n_{\alpha} n_{\beta} + \frac{1}{9} \eta_{\alpha\beta} C_{0^{+}n_{\chi}}^{\#2} +$ $\frac{1}{2} C_{1^{+}\alpha\beta}^{\#1} n_{\chi} + \frac{1}{3} C_{2^{+}\alpha\beta}^{\#1} n_{\chi} - \frac{1}{3} C_{2^{+}\alpha\beta}^{\#2} n_{\chi} - \frac{1}{9} \eta_{\alpha\beta} C_{0^{+}n_{\chi}}^{\#3} - \frac{1}{3} C_{1^{-}\beta}^{\#4} n_{\alpha} n_{\chi} - \frac{1}{15} C_{1^{-}\beta}^{\#2} n_{\alpha} n_{\chi} + \frac{2}{3} C_{1^{-}\beta}^{\#3} n_{\alpha} n_{\chi} +$ $\frac{1}{3} C_{1^{-}\beta}^{\#1} n_{\alpha} n_{\chi} + \frac{1}{6} C_{1^{-}\alpha}^{\#4} n_{\beta} n_{\chi} - \frac{1}{15} C_{1^{-}\alpha}^{\#2} n_{\beta} n_{\chi} - \frac{1}{3} C_{1^{-}\alpha}^{\#3} n_{\beta} n_{\chi} + \frac{1}{3} C_{1^{-}\alpha}^{\#1} n_{\beta} n_{\chi} - \frac{1}{3} C_{0^{+}n_{\alpha}}^{\#2} n_{\alpha} n_{\beta} n_{\chi} + C_{0^{+}n_{\alpha}}^{\#1} n_{\alpha} n_{\beta} n_{\chi}$	$J_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$C_{0^{+}}^{\#1}$	(None)	$C_{\alpha\beta\chi} n^{\alpha} n^{\beta} n^{\chi}$	$J_{0^{+}}^{\#1}$
$C_{0^{+}}^{\#2}$	(None)	$2 C_{\alpha}^{\beta} n^{\alpha} + C_{\alpha\beta}^{\beta} n^{\alpha} - 3 C_{\alpha\beta\chi} n^{\alpha} n^{\beta} n^{\chi}$	$J_{0^{+}}^{\#2}$
$C_{0^{+}}^{\#3}$	(None)	$-C_{\alpha}^{\beta} n^{\alpha} + C_{\alpha\beta}^{\beta} n^{\alpha}$	$J_{0^{+}}^{\#3}$
$C_{1^{+}\alpha\beta}^{\#1}$	$C_{1^{+}\beta\alpha}^{\#1} == -C_{1^{+}\alpha\beta}^{\#1}$	$C_{\alpha\beta\chi} n^{\chi} - C_{\beta\alpha\chi} n^{\chi} + C_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} - C_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - C_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta} + C_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$J_{1^{+}\alpha\beta}^{\#1}$
$C_{1^{-}\alpha}^{\#1}$	(None)	$2 C_{\alpha\beta\chi} n^{\beta} n^{\chi} + C_{\beta\alpha\chi} n^{\beta} n^{\chi} - 3 C_{\beta\chi\delta} n_{\alpha} n^{\beta} n^{\chi} n^{\delta}$	$J_{1^{-}\alpha}^{\#1}$
$C_{1^{-}\alpha}^{\#2}$	(None)	$2 C_{\alpha}^{\beta} n_{\beta} + C_{\alpha\beta}^{\beta} - 2 C_{\beta}^{\chi} n_{\chi} n_{\alpha} n^{\beta} - C_{\beta\chi}^{\chi} n_{\alpha} n^{\beta} - 2 C_{\alpha\beta\chi} n^{\beta} n^{\chi} - C_{\beta\alpha\chi} n^{\beta} n^{\chi} + 3 C_{\beta\chi\delta} n_{\alpha} n^{\beta} n^{\chi} n^{\delta}$	$J_{1^{-}\alpha}^{\#2}$
$C_{1^{-}\alpha}^{\#3}$	(None)	$-C_{\alpha\beta\chi} n^{\beta} n^{\chi} + C_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$J_{1^{-}\alpha}^{\#3}$
$C_{1^{-}\alpha}^{\#4}$	(None)	$-C_{\alpha}^{\beta} n_{\beta} + C_{\alpha\beta}^{\beta} + C_{\beta}^{\chi} n_{\chi} n_{\alpha} n^{\beta} - C_{\beta\chi}^{\chi} n_{\alpha} n^{\beta} + C_{\alpha\beta\chi} n^{\beta} n^{\chi} - C_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$J_{1^{-}\alpha}^{\#4}$
$C_{2^{+}\alpha\beta}^{\#1}$	$C_{2^{+}\beta\alpha}^{\#1} == C_{2^{+}\alpha\beta}^{\#1}$	$C_{\alpha\beta\chi} n^{\chi} + C_{\alpha\chi\beta} n^{\chi} + C_{\beta\alpha\chi} n^{\chi} - \frac{2}{3} \eta_{\alpha\beta} C_{\chi}^{\delta} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} C_{\chi\delta}^{\delta} n^{\chi} + \frac{2}{3} C_{\chi}^{\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{3} C_{\chi\delta}^{\delta} n_{\alpha} n_{\beta} n^{\chi} -$ $2 C_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} - C_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - 2 C_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta} - C_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta} + \eta_{\alpha\beta} C_{\chi\delta\epsilon} n^{\chi} n^{\delta} n^{\epsilon} + 2 C_{\chi\delta\epsilon} n_{\alpha} n_{\beta} n^{\chi} n^{\delta} n^{\epsilon}$	$J_{2^{+}\alpha\beta}^{\#1}$
$C_{2^{+}\alpha\beta}^{\#2}$	$C_{2^{+}\beta\alpha}^{\#2} == C_{2^{+}\alpha\beta}^{\#2}$	$-\frac{1}{2} C_{\alpha\beta\chi} n^{\chi} + C_{\alpha\chi\beta} n^{\chi} - \frac{1}{2} C_{\beta\alpha\chi} n^{\chi} + \frac{1}{3} \eta_{\alpha\beta} C_{\chi}^{\delta} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} C_{\chi\delta}^{\delta} n^{\chi} - \frac{1}{3} C_{\chi}^{\delta} n_{\alpha} n_{\beta} n^{\chi} +$ $\frac{1}{3} C_{\chi\delta}^{\delta} n_{\alpha} n_{\beta} n^{\chi} - \frac{1}{2} C_{\beta\chi\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} C_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - \frac{1}{2} C_{\alpha\chi\delta} n_{\beta} n^{\chi} n^{\delta} + \frac{1}{2} C_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$J_{2^{+}\alpha\beta}^{\#2}$
$C_{2^{-}\alpha\beta\chi}^{\#1}$	$C_{2^{-}\beta\alpha\chi}^{\#1} == -C_{2^{-}\alpha\beta\chi}^{\#1}$	$-\frac{2}{3} C_{\alpha\beta\chi} + \frac{1}{3} \eta_{\beta\chi} C_{\alpha}^{\delta} n_{\delta} + \frac{2}{3} C_{\beta\alpha\chi} - \frac{1}{3} \eta_{\alpha\chi} C_{\beta}^{\delta} n_{\delta} - \frac{1}{3} \eta_{\beta\chi} C_{\delta}^{\delta} n_{\alpha\delta} + \frac{1}{3} \eta_{\alpha\chi} C_{\delta}^{\delta} n_{\beta\delta} + \frac{1}{3} C_{\beta}^{\delta} n_{\alpha} n_{\chi} - \frac{1}{3} C_{\delta}^{\delta} n_{\alpha} n_{\chi} - \frac{1}{3} C_{\alpha}^{\delta} n_{\beta} n_{\chi} +$ $\frac{1}{3} C_{\delta}^{\delta} n_{\beta} n_{\chi} - \frac{2}{3} C_{\beta\delta\chi} n_{\alpha} n^{\delta} + \frac{2}{3} C_{\chi\beta\delta} n_{\alpha} n^{\delta} - \frac{1}{3} \eta_{\beta\chi} C_{\delta}^{\epsilon} n_{\alpha} n^{\delta} + \frac{1}{3} \eta_{\beta\chi} C_{\delta\epsilon}^{\epsilon} n_{\alpha} n^{\delta} + \frac{2}{3} C_{\alpha\delta\chi} n_{\beta} n^{\delta} - \frac{2}{3} C_{\chi\alpha\delta} n_{\beta} n^{\delta} +$ $\frac{1}{3} \eta_{\alpha\chi} C_{\delta}^{\epsilon} n_{\beta} n^{\delta} - \frac{1}{3} \eta_{\alpha\chi} C_{\delta\epsilon}^{\epsilon} n_{\beta} n^{\delta} + \frac{2}{3} C_{\alpha\beta\delta} n_{\chi} n^{\delta} - \frac{2}{3} C_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{3} \eta_{\beta\chi} C_{\alpha\delta\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{3} \eta_{\alpha\chi} C_{\beta\delta\epsilon} n^{\delta} n^{\epsilon} +$ $\frac{1}{3} \eta_{\beta\chi} C_{\delta\alpha\epsilon} n^{\delta} n^{\epsilon} - \frac{1}{3} \eta_{\alpha\chi} C_{\delta\beta\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{3} C_{\beta\delta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} - \frac{1}{3} C_{\delta\beta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} - \frac{1}{3} C_{\alpha\delta\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} + \frac{1}{3} C_{\delta\alpha\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$J_{2^{-}\alpha\beta\chi}^{\#1}$
$C_{3^{-}\alpha\beta\chi}^{\#1}$	$C_{3^{-}\alpha\chi\beta}^{\#1} == C_{3^{-}\alpha\beta\chi}^{\#1} \ \&\& \ C_{3^{-}\beta\alpha\chi}^{\#1} == C_{3^{-}\alpha\beta\chi}^{\#1} \ \&\& \ C_{3^{-}\beta\chi\alpha}^{\#1} == C_{3^{-}\alpha\beta\chi}^{\#1} \ \&\& \ C_{3^{-}\chi\alpha\beta}^{\#1} == C_{3^{-}\alpha\beta\chi}^{\#1} \ \&\& \ C_{3^{-}\chi\beta\alpha}^{\#1} == C_{3^{-}\alpha\beta\chi}^{\#1}$	$\frac{1}{3} C_{\alpha\beta\chi} + \frac{1}{3} C_{\alpha\chi\beta} - \frac{2}{15} \eta_{\beta\chi} C_{\alpha}^{\delta} n_{\delta} + \frac{1}{3} C_{\beta\alpha\chi} - \frac{2}{15} \eta_{\alpha\chi} C_{\beta}^{\delta} n_{\delta} - \frac{2}{15} \eta_{\alpha\beta} C_{\chi}^{\delta} n_{\delta} - \frac{1}{15} \eta_{\beta\chi} C_{\delta}^{\delta} n_{\alpha\delta} - \frac{1}{15} \eta_{\alpha\chi} C_{\delta}^{\delta} n_{\beta\delta} +$ $\frac{2}{15} C_{\chi}^{\delta} n_{\alpha} n_{\beta} + \frac{1}{15} C_{\chi\delta}^{\delta} n_{\alpha} n_{\beta} + \frac{2}{15} C_{\beta}^{\delta} n_{\alpha} n_{\chi} + \frac{1}{15} C_{\delta}^{\delta} n_{\alpha} n_{\chi} + \frac{2}{15} C_{\alpha}^{\delta} n_{\beta} n_{\chi} + \frac{1}{15} C_{\delta}^{\delta} n_{\beta} n_{\chi} - \frac{1}{3} C_{\beta\chi\delta} n_{\alpha} n^{\delta} -$ $\frac{1}{3} C_{\beta\delta\chi} n_{\alpha} n^{\delta} - \frac{1}{3} C_{\chi\beta\delta} n_{\alpha} n^{\delta} + \frac{2}{15} \eta_{\beta\chi} C_{\delta}^{\epsilon} n_{\alpha} n^{\delta} + \frac{1}{15} \eta_{\beta\chi} C_{\delta\epsilon}^{\epsilon} n_{\alpha} n^{\delta} - \frac{1}{3} C_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} C_{\alpha\delta\chi} n_{\beta} n^{\delta} - \frac{1}{3} C_{\chi\alpha\delta} n_{\beta} n^{\delta} +$ $\frac{2}{15} \eta_{\alpha\chi} C_{\delta}^{\epsilon} n_{\beta} n^{\delta} + \frac{1}{15} \eta_{\alpha\chi} C_{\delta\epsilon}^{\epsilon} n_{\beta} n^{\delta} - \frac{1}{3} C_{\alpha\beta\delta} n_{\chi} n^{\delta} - \frac{1}{3} C_{\alpha\delta\beta} n_{\chi} n^{\delta} - \frac{1}{3} C_{\beta\alpha\delta} n_{\chi} n^{\delta} + \frac{2}{15} \eta_{\alpha\beta} C_{\delta}^{\epsilon} n_{\chi} n^{\delta} +$ $\frac{1}{15} \eta_{\alpha\beta} C_{\delta\epsilon}^{\epsilon} n_{\chi} n^{\delta} - \frac{2}{5} C_{\delta}^{\epsilon} n_{\alpha} n_{\beta} n_{\chi} n^{\delta} - \frac{1}{5} C_{\delta\epsilon}^{\epsilon} n_{\alpha} n_{\beta} n_{\chi} n^{\delta} + \frac{2}{15} \eta_{\beta\chi} C_{\alpha\delta\epsilon} n^{\delta} n^{\epsilon} + \frac{2}{15} \eta_{\alpha\chi} C_{\beta\delta\epsilon} n^{\delta} n^{\epsilon} +$ $\frac{2}{15} \eta_{\alpha\beta} C_{\chi\delta\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{15} \eta_{\beta\chi} C_{\delta\alpha\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{15} \eta_{\alpha\chi} C_{\delta\beta\epsilon} n^{\delta} n^{\epsilon} + \frac{1}{15} \eta_{\alpha\beta} C_{\delta\chi\epsilon} n^{\delta} n^{\epsilon} + \frac{8}{15} C_{\chi\delta\epsilon} n_{\alpha} n_{\beta} n^{\delta} n^{\epsilon} +$ $\frac{4}{15} C_{\delta\chi\epsilon} n_{\alpha} n_{\beta} n^{\delta} n^{\epsilon} + \frac{8}{15} C_{\beta\delta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{4}{15} C_{\delta\beta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{8}{15} C_{\alpha\delta\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} + \frac{4}{15} C_{\delta\alpha\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} -$ $\frac{1}{5} \eta_{\beta\chi} C_{\delta\epsilon\phi} n_{\alpha} n^{\delta} n^{\epsilon} n^{\phi} - \frac{1}{5} \eta_{\alpha\chi} C_{\delta\epsilon\phi} n_{\beta} n^{\delta} n^{\epsilon} n^{\phi} - \frac{1}{5} \eta_{\alpha\beta} C_{\delta\epsilon\phi} n_{\chi} n^{\delta} n^{\epsilon} n^{\phi} - \frac{2}{5} C_{\delta\epsilon\phi} n_{\alpha} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon} n^{\phi}$	$J_{3^{-}\alpha\beta\chi}^{\#1}$